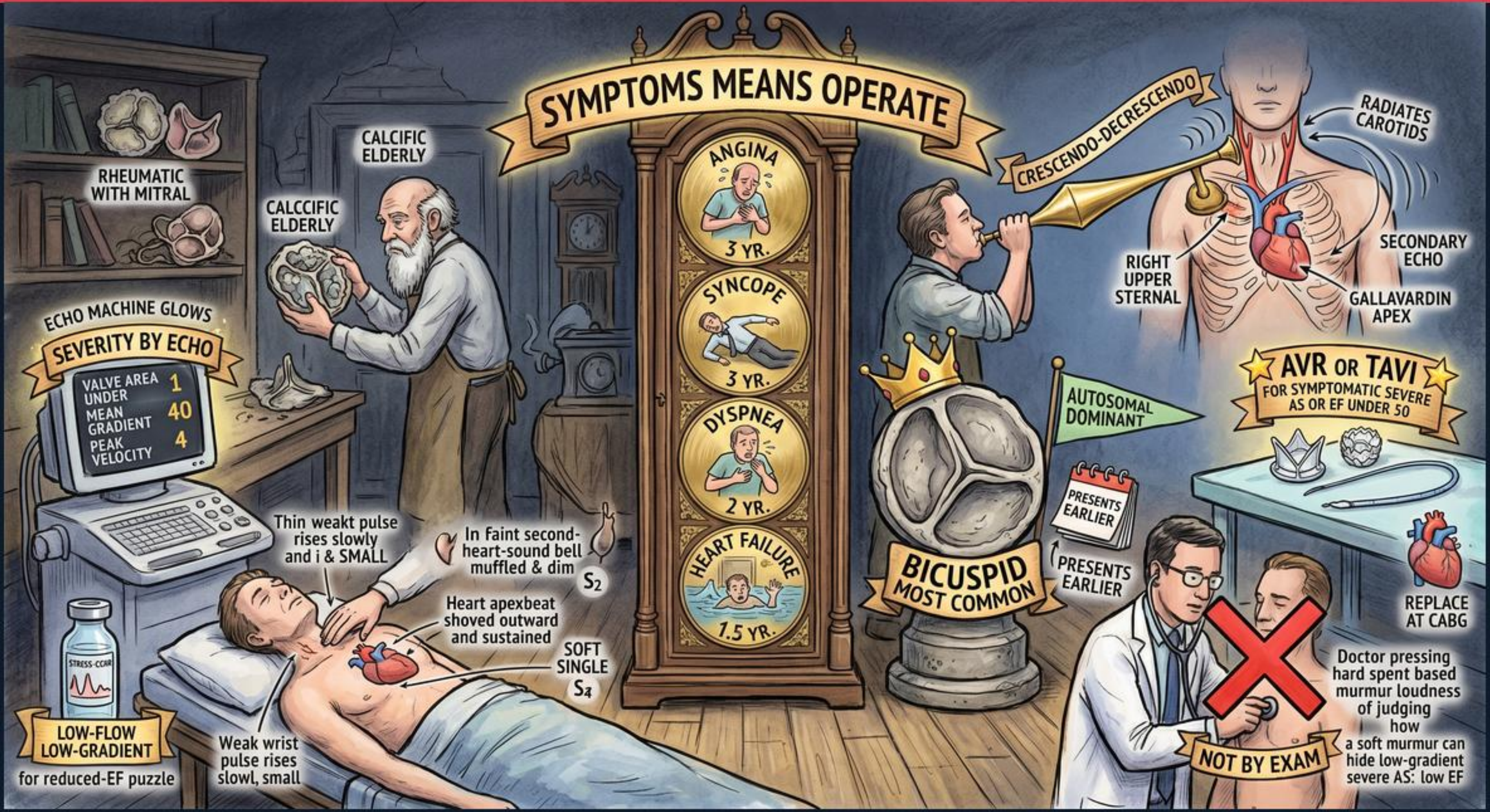
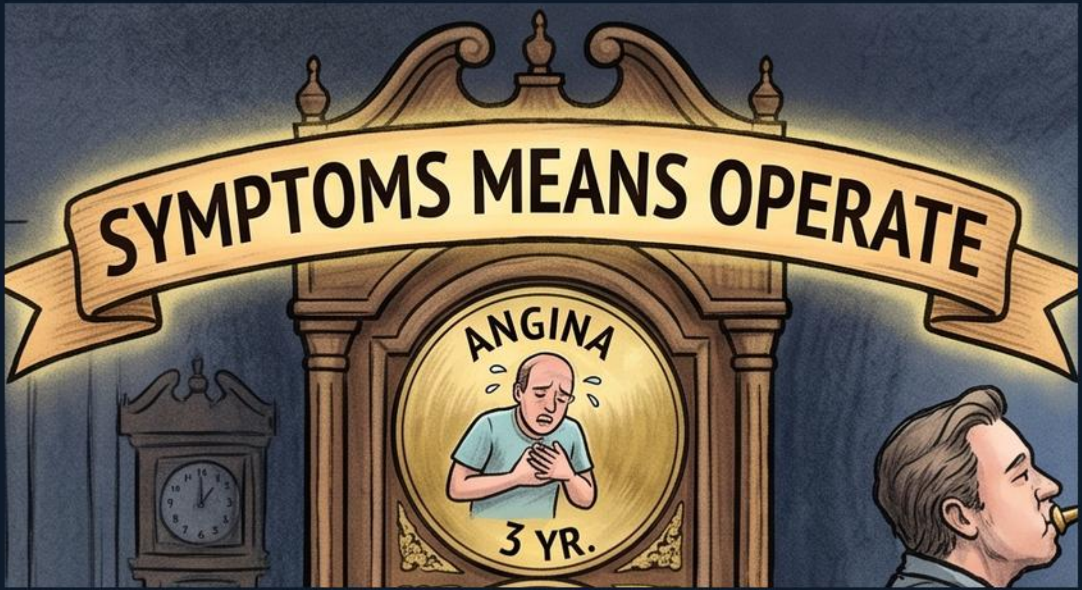
Valvular — Aortic Stenosis



16 symbols · sketch walkthrough



SYMPTOMS MEANS OPERATE (banner)

WHAT YOU SEE

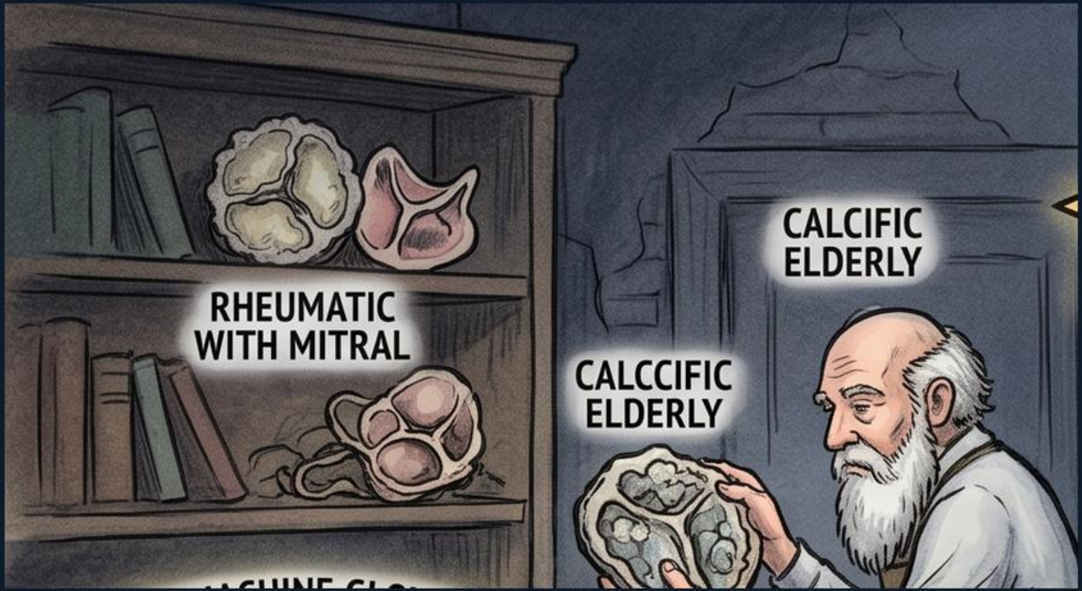
Top banner reading SYMPTOMS MEANS OPERATE.

WHAT IT ENCODES

Symptom onset in severe AS is the trigger to intervene — angina, syncope, or heart failure mark a steep drop in survival without surgery. Once symptomatic, severe AS carries a poor prognosis and AVR/TAVI is indicated (Class I). Symptoms convert a watch-and-wait lesion into a surgical one.

THE BOARD TRAP

Don't wait for all three symptoms — ANY symptom (angina, syncope, or dyspnea/HF) in severe AS means operate.



Rheumatic with mitral (shelf, top-left)

WHAT YOU SEE

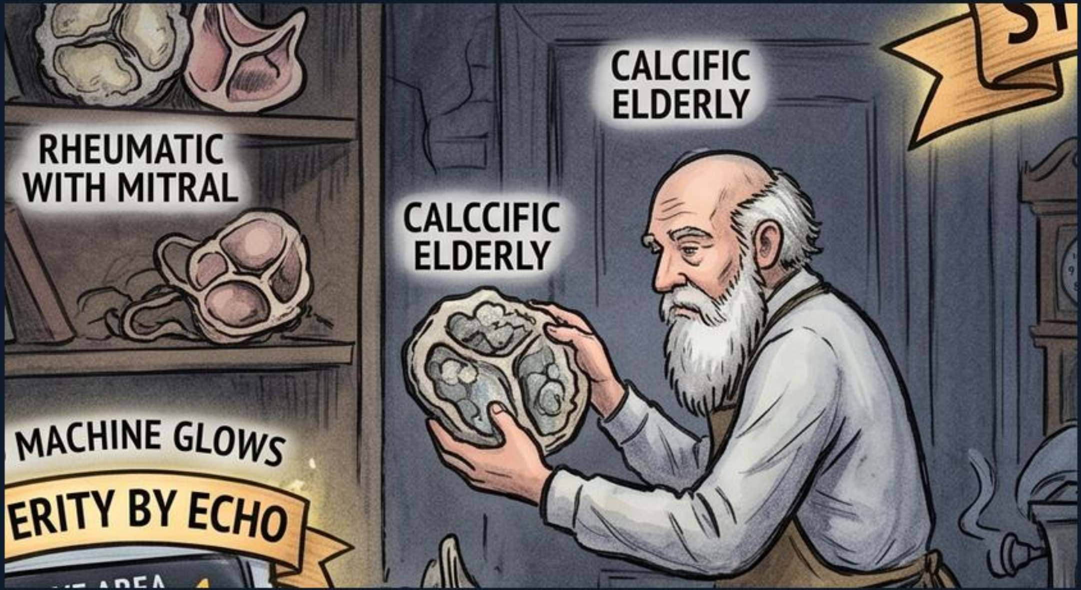
A book/shelf labeled RHEUMATIC WITH MITRAL.

WHAT IT ENCODES

Rheumatic aortic stenosis almost always accompanies rheumatic MITRAL valve disease — isolated rheumatic AS is rare. Finding AS plus mitral involvement points to a rheumatic etiology. This is the third major cause after bicuspid and calcific degeneration.

THE BOARD TRAP

Isolated AS without mitral disease is NOT typically rheumatic — think bicuspid (younger) or calcific (elderly) instead.



Calcific elderly man + valve

WHAT YOU SEE

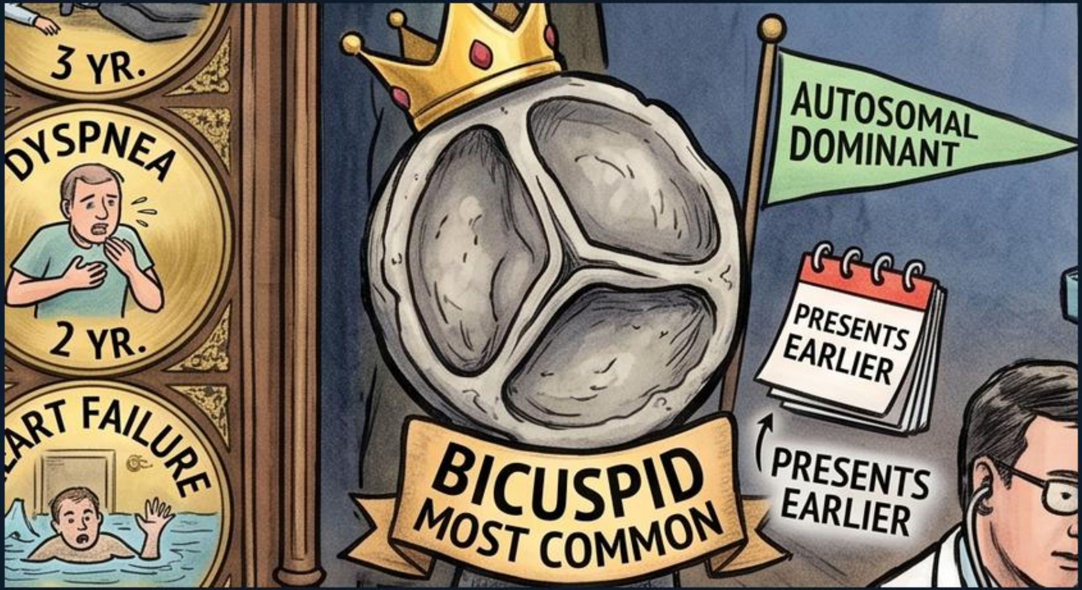
An old man holding a calcified valve, labeled CALCIFIC ELDERLY.

WHAT IT ENCODES

Calcific (degenerative) aortic stenosis is the leading cause in the ELDERLY, from age-related wear and calcium deposition on a previously normal trileaflet valve. It shares risk factors with atherosclerosis. This is the typical 70-80+ year-old AS patient.

THE BOARD TRAP

In an elderly patient, calcific degeneration — not rheumatic disease or congenital bicuspid — is the most likely cause of new AS.



BICUSPID MOST COMMON (two-leaflet valve)

WHAT YOU SEE

A large two-leaflet valve labeled BICUSPID MOST COMMON, crowned, with PRESENTS EARLIER flags.

WHAT IT ENCODES

Bicuspid aortic valve is the MOST COMMON congenital valve lesion, inherited autosomal dominant. It undergoes accelerated calcification and presents with AS 1-2 decades EARLIER than calcific degeneration (often 40s-60s). Suspect it in a younger adult with AS or an associated aortopathy.

THE BOARD TRAP

Bicuspid AS presents EARLIER than calcific AS — a 50-year-old with severe AS is far more likely bicuspid than degenerative.

Autosomal dominant (flag)

WHAT YOU SEE

A flag labeled AUTOSOMAL DOMINANT near the bicuspid valve.

WHAT IT ENCODES

Bicuspid aortic valve has autosomal dominant inheritance with variable penetrance, so first-degree relatives warrant screening. It is also associated with thoracic aortic dilation/aneurysm and coarctation. The genetics tie the valve to a broader aortopathy.

THE BOARD TRAP

Screen first-degree relatives and image the aorta — bicuspid disease is more than just a valve problem (aortopathy/coarctation).

Crescendo-decrescendo (arrow, top-right)

WHAT YOU SEE

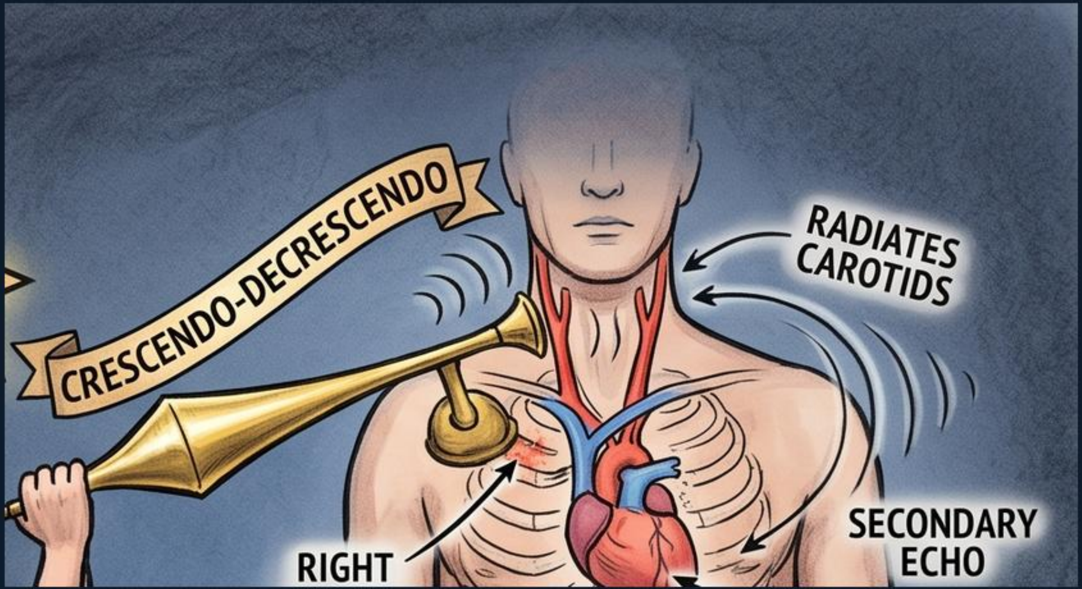
A crescendo-decrescendo diamond shape labeled near the right upper sternal area.

WHAT IT ENCODES

The AS murmur is a mid-systolic CRESCENDO-DECRESCENDO (ejection) murmur, loudest at the right upper sternal border. Later peaking of the murmur correlates with more severe stenosis. Its ejection shape distinguishes it from the holosystolic murmur of MR.

THE BOARD TRAP

A LATER-peaking murmur suggests more severe AS, but murmur timing/loudness still does NOT grade severity — echo does.



Radiates to carotids

WHAT YOU SEE

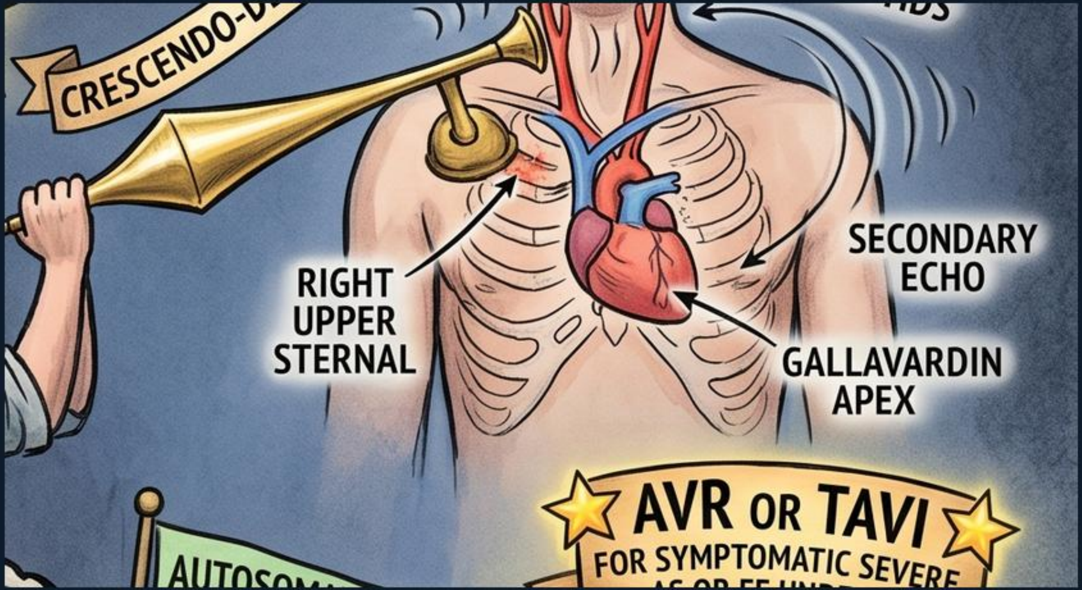
Arrow to the neck labeled RADIATES CAROTIDS.

WHAT IT ENCODES

The AS murmur classically radiates to the CAROTIDS. A palpable carotid thrill and delayed carotid upstroke accompany severe disease. Radiation to the neck helps separate AS from other systolic murmurs.

THE BOARD TRAP

Radiation to the carotids points to AS; the Gallavardin phenomenon radiates to the APEX and can mimic MR — same valve, different site.



Gallavardin → apex

WHAT YOU SEE

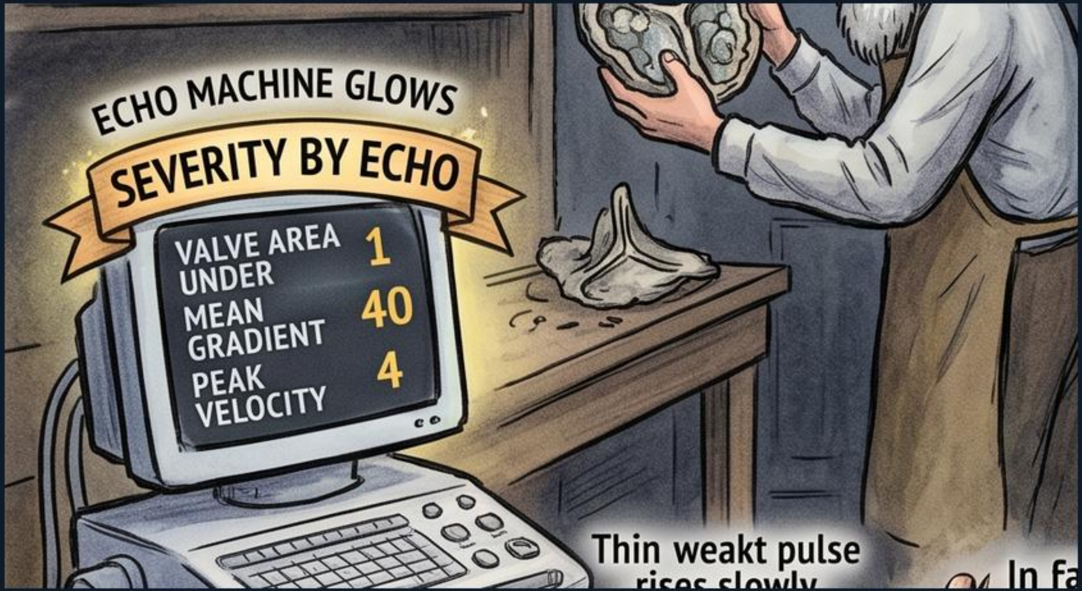
Label GALLAVARDIN APEX with a secondary radiation arrow to the apex.

WHAT IT ENCODES

The Gallavardin phenomenon is radiation of the AS murmur to the APEX, where its musical quality can mimic mitral regurgitation. It is still the aortic stenosis murmur, not a second lesion. Recognizing it prevents over-diagnosing MR.

THE BOARD TRAP

An apical systolic murmur in a patient with AS may be Gallavardin (AS radiating), not a separate MR — don't invent a second valve lesion.



Severity by echo panel (valve area 1.0, gradient 40, velocity 4)

WHAT YOU SEE

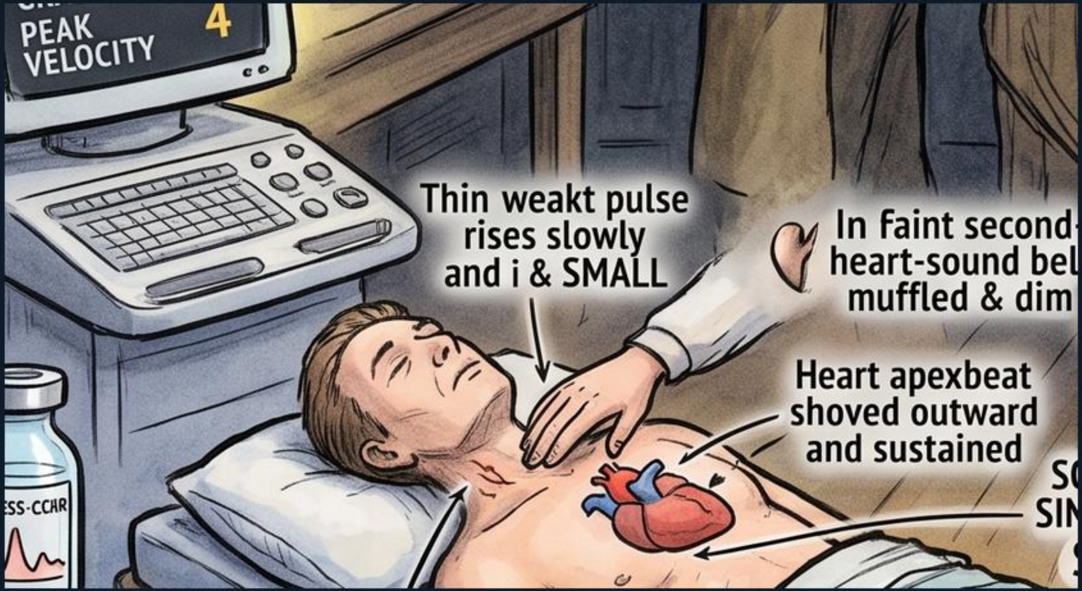
A glowing echo machine panel: VALVE AREA UNDER, MEAN GRADIENT, PEAK VELOCITY with 40 and 4.

WHAT IT ENCODES

Severity is defined by ECHO, not the exam: SEVERE AS = valve area <1.0 cm2, mean gradient >=40 mmHg, and peak jet velocity >=4 m/s. These three numbers anchor the diagnosis of severe disease. Get an echo before deciding on intervention.

THE BOARD TRAP

Grade severity by echo numbers (area <1.0, gradient >=40, velocity >=4) — NOT by how loud the murmur is on exam.



Pulsus parvus et tardus (weak slow pulse)

WHAT YOU SEE

A patient with text THIN WEAK PULSE RISES SLOWLY and a weak wrist pulse — slow and small.

WHAT IT ENCODES

Pulsus parvus et tardus is the carotid/peripheral pulse of severe AS: small in amplitude (parvus) and slow to rise (tardus). It reflects the fixed obstruction limiting ejection. A delayed, diminished carotid upstroke is a key bedside sign of severity.

THE BOARD TRAP

Parvus et tardus (weak AND late) is AS; a brisk, bounding/water-hammer pulse points to aortic REGURGITATION instead.

Soft / single / paradoxically split S2

WHAT YOU SEE

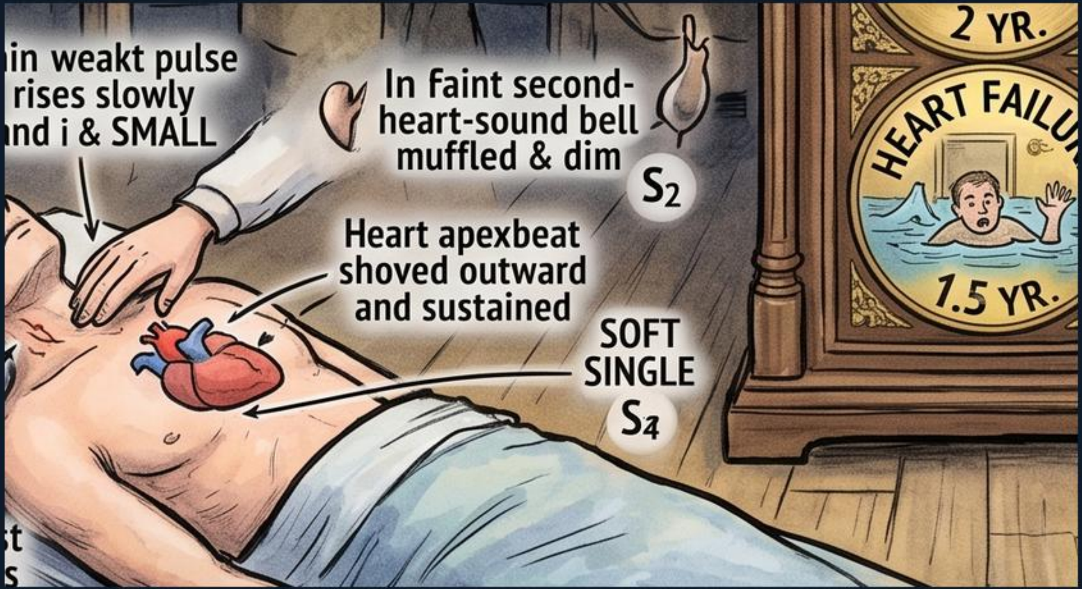
Label SOFT SINGLE S2 by the chest exam figure.

WHAT IT ENCODES

In severe AS the aortic component of S2 becomes soft or inaudible, giving a SOFT or SINGLE S2; with very severe disease A2 is so delayed it produces PARADOXICAL splitting (split on expiration). A soft/single S2 is a marker of severe AS. The immobile calcified valve fails to generate a crisp A2.

THE BOARD TRAP

A soft or single S2 signals SEVERE AS — a normally split S2 argues against severe stenosis.



S4 / sustained displaced apex

WHAT YOU SEE

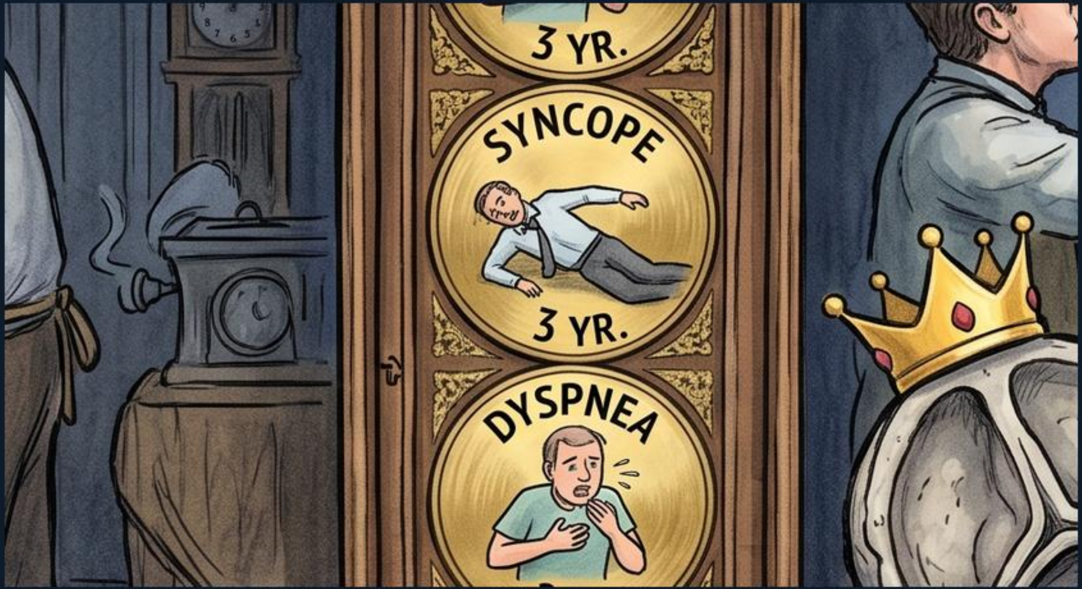
Label HEART APEXBEAT SHOVED OUTWARD AND SUSTAINED, with bell muffled S4 cues.

WHAT IT ENCODES

Pressure-overload LV hypertrophy in AS produces an S4 (atrial kick into a stiff ventricle) and a SUSTAINED, sometimes displaced apical impulse. These reflect the hypertrophied, non-compliant LV. They support but do not grade severity.

THE BOARD TRAP

An S4 and sustained apex indicate LVH from AS but do NOT quantify severity — only echo grades the lesion.



Symptom→survival ladder (angina/syncope/dyspnea/HF, clock)

WHAT YOU SEE

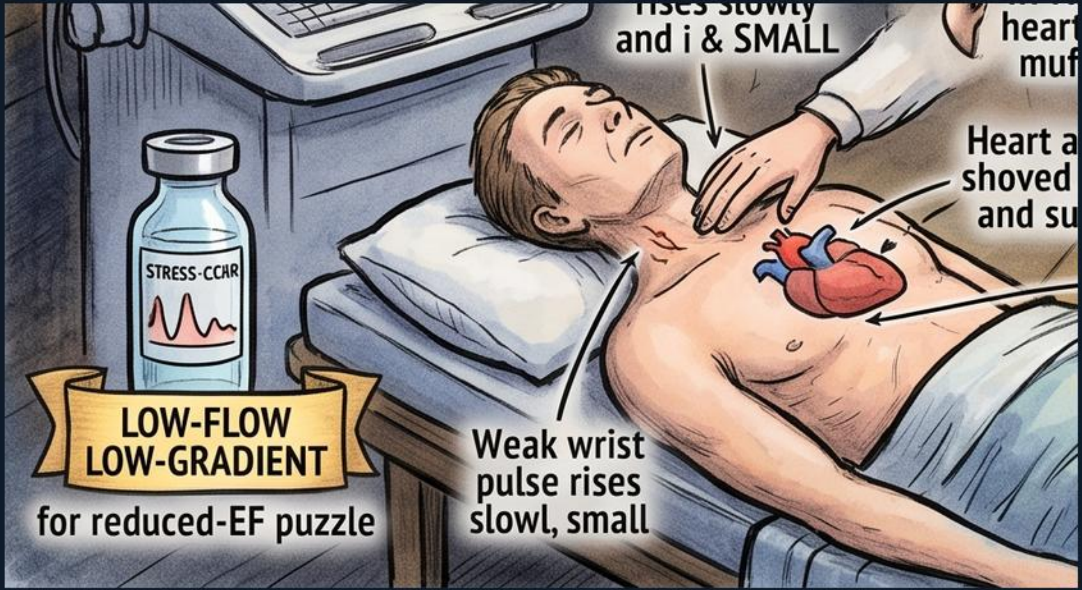
A grandfather clock listing ANGINA 3 YR, SYNCOPE 3 YR, DYSPNEA 2 YR, HEART FAILURE 1.5 YR.

WHAT IT ENCODES

Symptom onset marks a sharp prognostic decline without surgery: angina ~3 yr, syncope ~3 yr, dyspnea ~2 yr, and heart failure ~1.5 yr median survival. The sicker the symptom, the shorter the time. Any of these in severe AS mandates intervention.

THE BOARD TRAP

Heart failure carries the WORST prognosis (~1.5 yr) — don't reassure a dyspneic/HF patient; symptom onset of any kind means operate.



Low-flow/low-gradient → dobutamine stress echo

WHAT YOU SEE

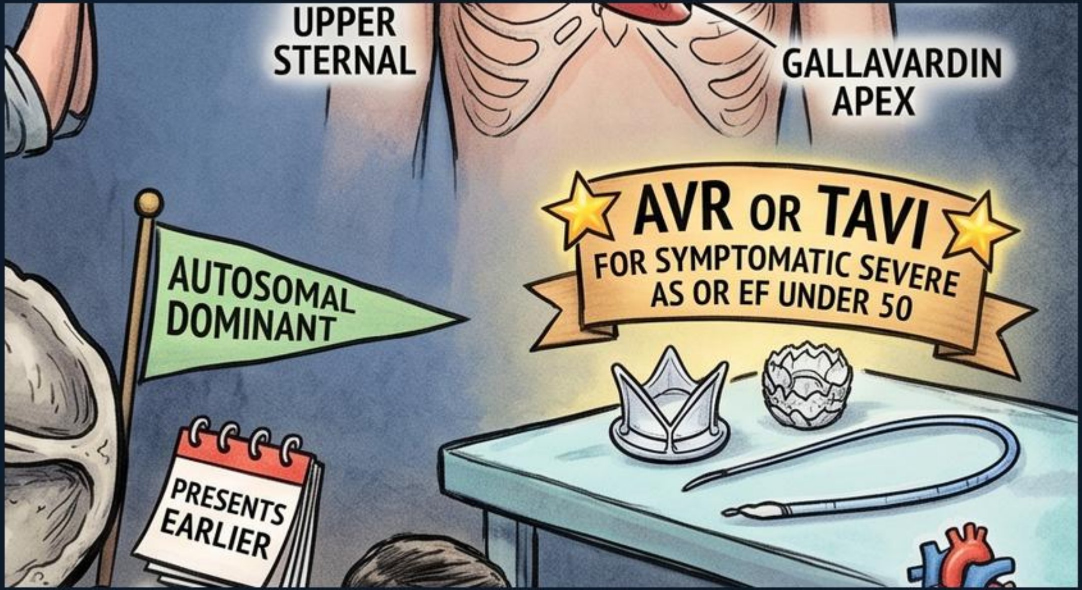
A STRESS ECHO bottle labeled LOW-FLOW LOW-GRADIENT for reduced-EF puzzle.

WHAT IT ENCODES

When the valve area says severe but the gradient/velocity are low because EF is reduced (low-flow, low-gradient AS), use DOBUTAMINE STRESS ECHO. Augmenting flow distinguishes true severe AS (gradient rises, area stays small) from pseudo-severe AS (area opens up). This resolves the discordant-numbers puzzle.

THE BOARD TRAP

Low gradient does NOT rule out severe AS when EF is low — do a dobutamine stress echo before excluding severe disease.



AVR or TAVI / replace at CABG

WHAT YOU SEE

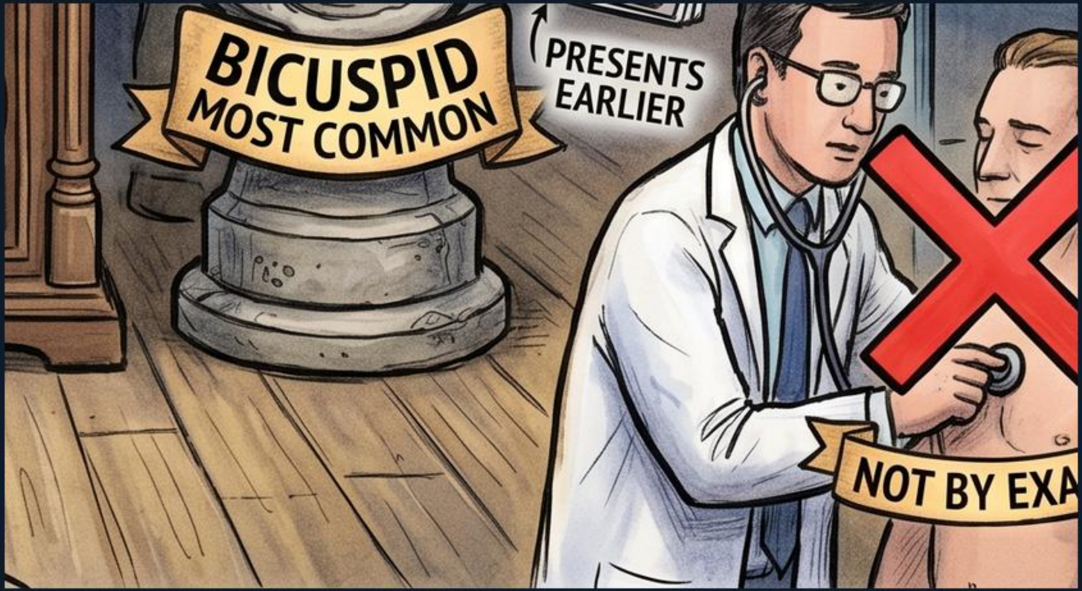
Banner AVR OR TAVI FOR SYMPTOMATIC SEVERE AS OR EF UNDER 50, with REPLACE AT CABG.

WHAT IT ENCODES

Class I intervention (surgical AVR or TAVI): symptomatic severe AS, OR asymptomatic severe AS with EF <50%. Also replace the valve at the time of CABG or other cardiac surgery if severe AS is present. The choice between AVR and TAVI depends on surgical risk and anatomy.

THE BOARD TRAP

Operate on severe AS even if asymptomatic when EF <50% — and replace it during CABG; don't defer just because symptoms are absent.



NOT BY EXAM — soft murmur trap (red X)

WHAT YOU SEE

A doctor pressing hard with a red X, labeled NOT BY EXAM — a soft murmur can hide low-gradient severe AS.

WHAT IT ENCODES

Severity is NOT judged by murmur loudness. A SOFT murmur can represent low-gradient severe AS with a low EF, because a weak ventricle cannot generate a loud turbulent jet. Always confirm with echo, and consider dobutamine stress echo if EF is low. Loudness misleads in exactly the sickest patients.

THE BOARD TRAP

A SOFT murmur does NOT mean mild AS — with low EF it can be severe low-gradient AS; grade by echo, never by exam intensity.

Summary — every symbol

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2. Rheumatic with mitral (shelf, top-left)

Rheumatic aortic stenosis almost always accompanies rheumatic MITRAL valve disease — isolated rheumatic AS is rare

3. Calcific elderly man + valve

Calcific (degenerative) aortic stenosis is the leading cause in the ELDERLY, from age-related wear and calcium deposition on a previously normal trileaflet valve

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